

Niti Kumari
Bengaluru, Karnataka
India, 560068
iamnitikumari@outlook.com

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To,
The Human Resource and Scientific Community,
Sri Shankara National Centre for Cancer Prevention and Research
Bengaluru, Karnataka

Subject: Inquiry regarding Scientific roles at Sri Shankara National Centre for Cancer Prevention and Research.

Dear Members of the Scientific Committee,

I am writing to express my interest in joining the Scientific Team at Sri Shankara National Centre for Cancer Prevention and Research (SSCHRC), Bengaluru. I would like to submit my profile for your consideration for any upcoming suitable opportunities. My academic qualifications include a Master's in Biotechnology from Panjab University and a Ph.D. in Biochemistry from the Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh, India. Additionally, I have 6.5 years of experience as a Postdoctoral Research Associate at Creighton University in Omaha, NE, USA.

My research is focused on understanding the molecular aspects of human pathophysiology, with a primary emphasis on cancer biology. My journey into cancer research started during my doctoral training at PGIMER, Chandigarh as a collaborative project, where I contributed to understanding the epigenetics of Renal Cell Carcinoma. During my postdoctoral training in the USA, I focused on understanding how tumors resist hostile nutritional and therapeutic stress environments and found an important role of MAPKAPK2 and ERK crosstalk. Additionally, I identified that nutritional supplements presumed to prevent aging-associated cancers, specifically skin cancer, might paradoxically enhance them. Simultaneously, I developed an *in vivo* model of metastatic colonization using CRISPR-Cas9 to identify which genes support metastatic colonization in Hepatocellular Carcinoma.

My expertise spans multiple dimensions of the field, including epigenetics, genetics, transcriptomics, proteomics, and multi-omics. My work is published in several reputed journals, including *PNAS*, *Seminars in Cancer Biology*, *Mediators of Inflammation*, *Human Mutation*, and *Scientific Reports*.

I am seeking a professional role where I can utilize my technical and analytical skills towards the advancement of high-impact research. I have attached my detailed *curriculum vitae* and a brief statement of my research interests for your review. I look forward to the opportunity to share my research journey so far and discuss my future research goals in detail at SSCHRC.

Sincerely,



Niti Kumari

Research Interest at Sri Shankara National Centre for Cancer Prevention and Research

- 1. Biomarker Discovery:** I am interested in identifying a molecule or a set of molecules to help stratification of patients for personalized medicine and evaluating recurrence.
- 2. Protein Homeostasis in Cancer:** Protein homeostasis is essential for cellular function and involves a complex array of processes, involving protein synthesis, folding, transportation to various subcellular compartments and degradation. Turnover of proteins is regulated by the ubiquitin-proteasome system (UPS) and the autophagy-lysosomal system. In ubiquitin proteasome system, several E3 ubiquitin ligases are reported to be dysregulated in cancer. Since, E3 ubiquitin ligases target specific substrates, I am interested in identifying novel E3 ubiquitin therapeutic targets responsible for disease heterogeneity.
- 3. NAD⁺ homeostasis in Cancer:** NAD⁺ is an important cofactor for various cellular reactions, including redox reactions, DNA repair and cell signaling. While decline of NAD⁺ is associated with aging and aging-associated diseases, its abundance regulates cancer progression and Inflammation. I am interested in understanding how NAD⁺ imbalance regulates aging as well as cancer.
- 4. Aging as driver of Cancer:** Aging is paramount factor for aging-and age-associated diseases, including cancer. Aging is associated with increase in incidence in cancer. What cellular changes makes human body vulnerable to cancer is an important step towards managing cancer incidence and increasing health span.

Niti Kumari, Ph.D.

Bengaluru, KA, India

Email: iamnitikumari@outlook.com

+91-8427099324

Postdoctoral Research Associate

Scholar Profile: <https://scholar.google.co.in/0rCB3kIAAAAJ>

LinkedIn ID: 8a7a5116b

Biomedical Sciences
School of Medicine
Creighton University
Omaha, Nebraska
68178, United States

PROFESSIONAL SUMMARY

I am a research professional with a **Ph.D. in Biochemistry** from the Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh (NIRF Ranking #2) and 6.5 years of **postdoctoral research experience at Creighton University**, one of the **top prestigious medical universities in Omaha, USA**. My research addresses the molecular mechanisms of human pathophysiology, with a primary emphasis on **oncogenesis, therapy resistance, metastasis, and cellular aging**.

My expertise spans multiple dimensions of the field, including **epigenetics, genetics, transcriptomics, and proteomics**. I utilize diverse disease models, including **in vitro, in vivo, and organoid systems**, to study cancer progression, metastasis and therapy resistance. My post-doctoral work identified p38/MAPKAPK2 and ERK signaling crosstalk as a critical driver of **therapy resistance in colorectal carcinoma**. Furthermore, I uncovered a **paradoxical role of NAD⁺ boosting** (a widely studied anti-aging intervention) in actively enhancing skin tumor formation. I also established an in vivo CRISPR based mouse model of metastatic colonization.

My work is published in several reputed journals, including *PNAS*, *Seminars in Cancer Biology*, *Mediators of Inflammation*, *Human Mutation*, *Scientific Reports*, and *Cancer Research* (Abstracts) with **two additional manuscripts currently under revision**. Beyond research, I have **peer reviewed** for journals such as Cellular Oncology and Frontiers in Oncology, and have **mentored** junior researchers in technical skill and experimental research.

My broader **research vision** centers on decoding the adaptive mechanisms employed by **cancer cells to survive therapeutic or nutritional stress**, the context-dependent roles of signaling pathways impacting tumor sensitivity versus resistance, and the **regulators of metastasis**. Additionally, I am interested in understanding the **molecular changes during aging** that create a permissive environment for cancer emergence. By integrating **high-throughput multi-omics data and bioinformatics** with traditional molecular biology, I aim to map genomic, transcriptomic, metabolomic, and proteomic dysregulations to identify novel disease biomarkers and therapeutic targets.

I am currently seeking a professional faculty or research role in an academic setting where I can utilize my technical and analytical skills toward the advancement of high-impact research in India.

RESEARCH EXPERIENCE

1. School of Medicine, Creighton University, Omaha, NE, USA

Position: Postdoctoral Research Associate

Duration: March 2019-August 2025

Mentor: Dr. Brian North

Responsibilities: Designing research study, Conducting research, Data Acquisition, Data Analysis and Interpretation, Data presentation to various scientific committees, Manuscript writing, Mentoring juniors, Grant writing and Collaboration, Laboratory Management.

Research Projects:

- Context-dependent role of p38/MAPKAPK2 and ERK signalling in colorectal cancer cells to survive under stress conditions, such as nutrient restriction and therapeutic stress.
- Role of NAD⁺ in shaping genetic and transcriptomic landscape in age-associated Skin cancer.
- Establishment of mouse model of liver cancer metastatic colonization using CRISPR technology.

2. Postgraduate Institute of Medical Education and Research, Chandigarh, India

Position: Doctoral Fellow (PhD)

Duration: Jan 2013- Feb 2019

Mentor: Dr. Rajendra Prasad

Doctoral Thesis: To establish a correlation between genotypes of Cu-*ATP7B* gene and its haplotypes, modifier genes *COMMD1* & *ATOX1*, Arg72Pro of *TP53* and phenotypes of Wilson disease.

Other lab Project(s): Understanding the role of H3K4me changes in Clear Cell Renal Cell Carcinoma.

3. Postgraduate Institute of Medical Education and Research, Chandigarh, India

Position: Research Assistant

Duration: Apr 2012- Dec 2012

Mentor: Dr. Rajendra Prasad

Project: Identification and Molecular Characterization of Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) mutations and microdeletions in human Y chromosome AZF candidate gene in male infertility: Diagnostic and Genetic implications

EDUCATION

2009-2011 Master of Science, Biotechnology
Panjab University, Chandigarh, India

2006-2009 Bachelor of Science, Biotechnology (Hons.)
Panjab University, Chandigarh, India

SCHOLARSHIP

2014-2017 Indian Council of Medical Research (ICMR)- Senior Research Fellowship

2012-2014 Indian Council of Medical Research (ICMR)- Junior Research Fellowship

2011 GATE (Biotechnology) with AIR 24

WORKSHOP/SYMPOSIUM/CONFERENCE

- American Association for Cancer Research (AACR), 2025 (Poster presentation).
- 44th Annual Meeting of the Indian Association for Cancer Research (IACR), 2025 (Poster presentation).
- Lynch Cancer Research Symposium 2024, conducted by Creighton University, Omaha, USA.
- Midlands society of physiological sciences (MSPS), 2024, conducted by University of South Dakota, Vermillion, SD, USA (Poster presentation).
- Annual Creighton University research week, 2024, conducted by Creighton University, Omaha, USA.
- Annual Creighton University research week, 2023, conducted by Creighton University, Omaha, USA.
- Annual Creighton University research week, 2022, conducted by Creighton University, Omaha, USA.

- Annual Creighton University research week, 2021, conducted by Creighton University, Omaha, USA.
- Annual Creighton University research week, 2020, conducted by Creighton University, Omaha, USA.
- ACBICON-2015 pre-conference workshop on “Applications of GC-MS and Tandem-Mass in inborn errors of metabolism and Flow Cytometry in Clinical laboratory” organized by the department of biochemistry, PGIMER on Nov 25, 2015.
- Workshop on “Fluorescence in situ Hybridization (FISH) Technology” conducted by Government Medical College Hospital (GMCH), Chandigarh from July 16- July 17, 2014.
- CME on “Advances in Clinical Biochemistry” organized by Department of Biochemistry, PGIMER, Chandigarh from March 23-24, 2013.
- National Symposium on “Animals in Biomedical Research” organized by Department of Biochemistry, PGIMER, Chandigarh on April 11, 2013.
- CME on “Molecular Diagnosis of Genetic Diseases” organized by Department of Biochemistry, PGIMER, Chandigarh from December 22-23, 2012.
- National Symposium on "Biotechnology: Innovations and Challenges" sponsored by the Department of Biotechnology, Govt. of India and organized by Department of Biotechnology & Biochemistry, GGDSD College Chandigarh on February 6th, 2010.
- National Symposium on "Advances in Biotechnology" sponsored by the Department of Science & Technology, Chandigarh and organized by the Department of Biotechnology, GGDSD College Chandigarh on February 3rd, 2007.

INVITED TALKS

- International seminar on cancer research and molecular diagnostics organized by department of biochemistry, Daulat Ram College Delhi University on Jan 30, 2025.

AWARDS

- Lynch Cancer Research Center Trainee Award by Creighton University, Omaha for AACR 2025 Conference.

MEMBERSHIP

- American Association for Cancer Research (AACR)
- Indian Association of Cancer Research (IACR)

PUBLICATIONS

Articles

1. **Kumari N**, Chen X, Baldwin AM, Clemons KI, El-Harakeh M, Calisto LE, Kumar B, Zhang Q, Min J, Xiao B, Singh B, Wang B, & North BJ, Metabolic stress conditions dictate MAPKAPK2-dependent efficiency of MEK1/2 inhibition in colorectal carcinoma, Proc. Natl. Acad. Sci. U.S.A. 123 (7) e2505331123, <https://doi.org/10.1073/pnas.2505331123>

2. Pun R, **Kumari N**, Monieb RH, Wagh S, North BJ. BubR1 and SIRT2: Insights into Aneuploidy, Aging, and Cancer. *Semin Cancer Biol.* 2024 Oct 26:S1044-579X(24)00083-X. doi: 10.1016/j.semcancer.2024.10.005. Epub ahead of print. PMID: 39490401.
<https://www.sciencedirect.com/science/article/pii/S1044579X2400083X?via%3Dihub>. **JIF 15.7**
3. **Kumari N**, Reabroi S, North BJ. Unraveling the Molecular Nexus between GPCRs, ERS, and EMT. *Mediators Inflamm.* 2021 Mar 2;2021:6655417. doi: 10.1155/2021/6655417. PMID: 33746610; PMCID: PMC7943314.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7943314/>. **JIF 4.2**
4. Kumar A, Nallabelli N, Sharma U, **Kumari N**, Singh SK, Kakkar N, Prasad R. In vitro evidence of NLRP3 inflammasome regulation by histone demethylase LSD2 in renal cancer: a pilot study. *Mol Biol Rep.* 2020 Sep;47(9):7273-7276. doi: 10.1007/s11033-020-05692-0. Epub 2020 Aug 4. PMID: 32754863.
<https://link.springer.com/article/10.1007/s11033-020-05692-0>. **JIF 2.8**
5. Kumar A, **Kumari N**, Nallabelli N, Sharma U, Rai A, Singh SK, Kakkar N, Prasad R. Expression profile of H3K4 demethylases with their clinical and pathological correlation in patients with clear cell renal cell carcinoma. *Gene.* 2020 May 20;739:144498. doi: 10.1016/j.gene.2020.144498. Epub 2020 Feb 22. PMID: 32097694.
<https://www.sciencedirect.com/science/article/pii/S0378111920301670?via%3Dihub>. **JIF 2.4**
6. Kumar A, **Kumari N**, Sharma U, Kaushal K, Singh SK, Kakkar N. Reduction in H3K4me code due to aberrant expression of methyltransferases and demethylases in renal cell carcinoma: prognostic and therapeutic implications. *Scientific Reports.* 2019, 9:8189. PMID: 31160694. DOI: 10.1038/s41598-019-44733-y.
<https://www.nature.com/articles/s41598-019-44733-y>. **JIF 3.9**
7. **Kumari N**, Kumar A, Thapa BR, Modi M, Prasad R. In-silico analysis of novel Gly14Ser variant of ATOX1 gene: Plausible role in modulating ATOX1-ATP7B interaction. *Molecular Biology Reports.* 2019, 46: 3307-3313. PMID: 30980273. DOI: 10.1007/s11033-019-04791-x <https://link.springer.com/article/10.1007%2Fs11033-019-04791-x> **JIF 2.8**
8. Kumar A, **Kumari N**, Nallabelli N, Prasad R. Pathogenic and Therapeutic Role of H3K4 Family of Methylases and Demethylases in Cancers. *Indian Journal of Clinical Biochemistry.* 2019, 34:123-132. PMID: 31092985. DOI: 10.1007/s12291-019-00828-x. <https://link.springer.com/article/10.1007/s12291-019-00828-x> . **JIF 1.6**
9. **Kumari N**, Kumar A, Thapa BR, Modi M, Pal A, Prasad R. Characterization of mutation spectrum and identification of novel mutations in ATP7B gene from a cohort of Wilson disease patients: Functional and Therapeutic Implications. *Human Mutation*, 2018, 39: 1–16. PMID: 30120852. doi: 10.1002/humu.23614.
<https://onlinelibrary.wiley.com/doi/abs/10.1002/humu.23614>. **JIF 3.7**
10. Kumar A, **Kumari N**, Rai A, Singh SK, Kakkar N, Prasad R. Expression and clinical significance of COMPASS family of histone methyltransferases in clear cell renal carcinoma. *Gene*, 2018, 674:31-36. PMID: 29944950. DOI: 10.1016/j.gene.2018.06.063. <https://www.sciencedirect.com/science/article/pii/S0378111918307194?via%3Dihub>. **JIF 2.4**
11. Kumar A, **Kumari N**, Gupta V, Prasad R. Renal Cell Carcinoma: Molecular Aspects. *Indian Journal of Clinical Biochemistry.* 2017, 33: 246-254. PMID: 30072823. DOI: 10.1007/s12291-017-0713-y
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6052717/>. **JIF 1.6**

Conference (Abstract)

- **Niti Kumari**, Brian J. North. NMN supplementation enhances proliferation and aggressiveness in UV-induced skin cancer [abstract]. In: Proceedings of the American Association for Cancer Research Annual Meeting 2025; Part 1 (Regular Abstracts); 2025 Apr 25-30; Chicago, IL. Philadelphia (PA): AACR; Cancer Res 2025;85(8_Suppl_1):Abstract nr 4168.
- **Niti Kumari**, Xu Chen, Amber M. Baldwin, Kristin I. Clemons, Mohammad Alharakeh, Lilian Calisto, Balawant Kumar, Qiaoqiao Zhang, Jiang Min, Bin Xiao, Amar B. Singh, Bin Wang, Brian J. North. Metabolic stress conditions dictate MAPKAPK2-dependent efficiency of MEK1/2 inhibition in colorectal carcinoma [abstract]. In: Proceedings of the American Association for Cancer Research Annual Meeting 2026; Part 1 (Regular Abstracts); 2026 Apr 17-22; San Diego, CA. Philadelphia (PA): AACR; Cancer Res 2026;86(7 Suppl):Abstract nr 5981.

Book Chapters

1. Thakur, R., **Kumari**, N., Mishra, D. P. (2021). Isolation and characterization of cancer stem cells. In Misra, G., Rajawat, J. (Eds.), Chapter 6 of Protocol Handbook for Cancer Biology (pp. 87-105). Academic Press. DOI: 10.1016/B978-0-323-90006-5.00007-0.
2. Thakur, R., **Kumari**, N. (2025). Targeting Metabolic Reprogramming: A New Frontier in Cancer Therapy. In: Ramteke, A.S., Ramakrishnan, S.K., Agarwal, R. (eds) Molecular Targets in Cancer Therapy. Springer, Singapore. https://doi.org/10.1007/978-981-96-7336-0_9.

Manuscript under revision

1. **Kumari N**, Siref AB, Clemons KI, Kuwong PT, Swedek MS, Kujur A, Pun R, Reabroi S, Bloom C, Hansen LA, North BJ. Role of NAD⁺ in skin tumorigenesis. **Under Revision in Cancer Letters.**

References

Upon Request